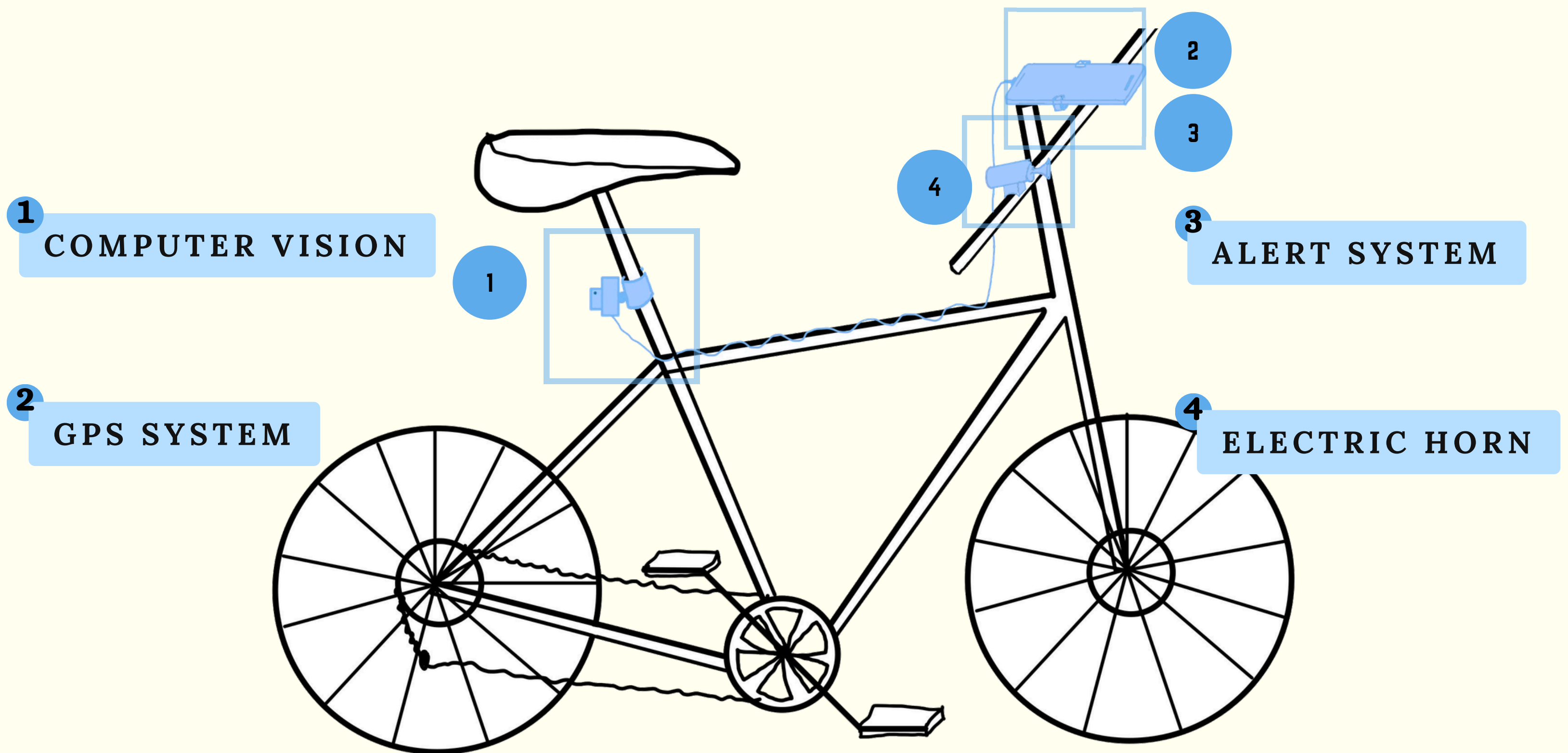


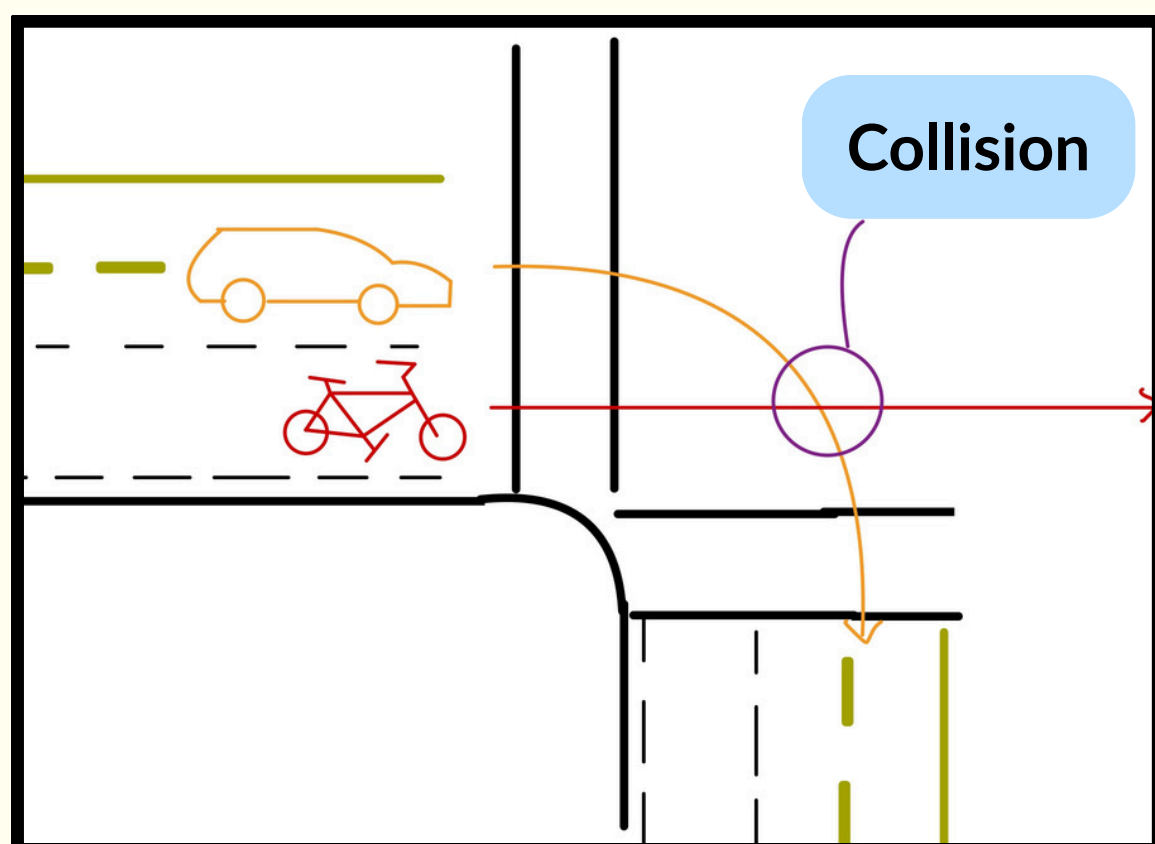
SOTERIA:

BIKE COLLISION DETECTION AND PREVENTION SYSTEM



This design aims to improve the awareness of drivers and cyclists in responding to a potential collision.

Using **computer vision**, it detects cars from the cyclist's blind-spot. To save battery, it only activates at intersections, which is achieved by a **GPS system**. Then, it activates an **alert system** on the user's phone. The cyclist can then respond with an **electric horn**.



THE OPPORTUNITY: RIGHT HOOK COLLISIONS

A collision in which a car, in a biker's blind-spot, turns right. Unable to see each other, they crash.

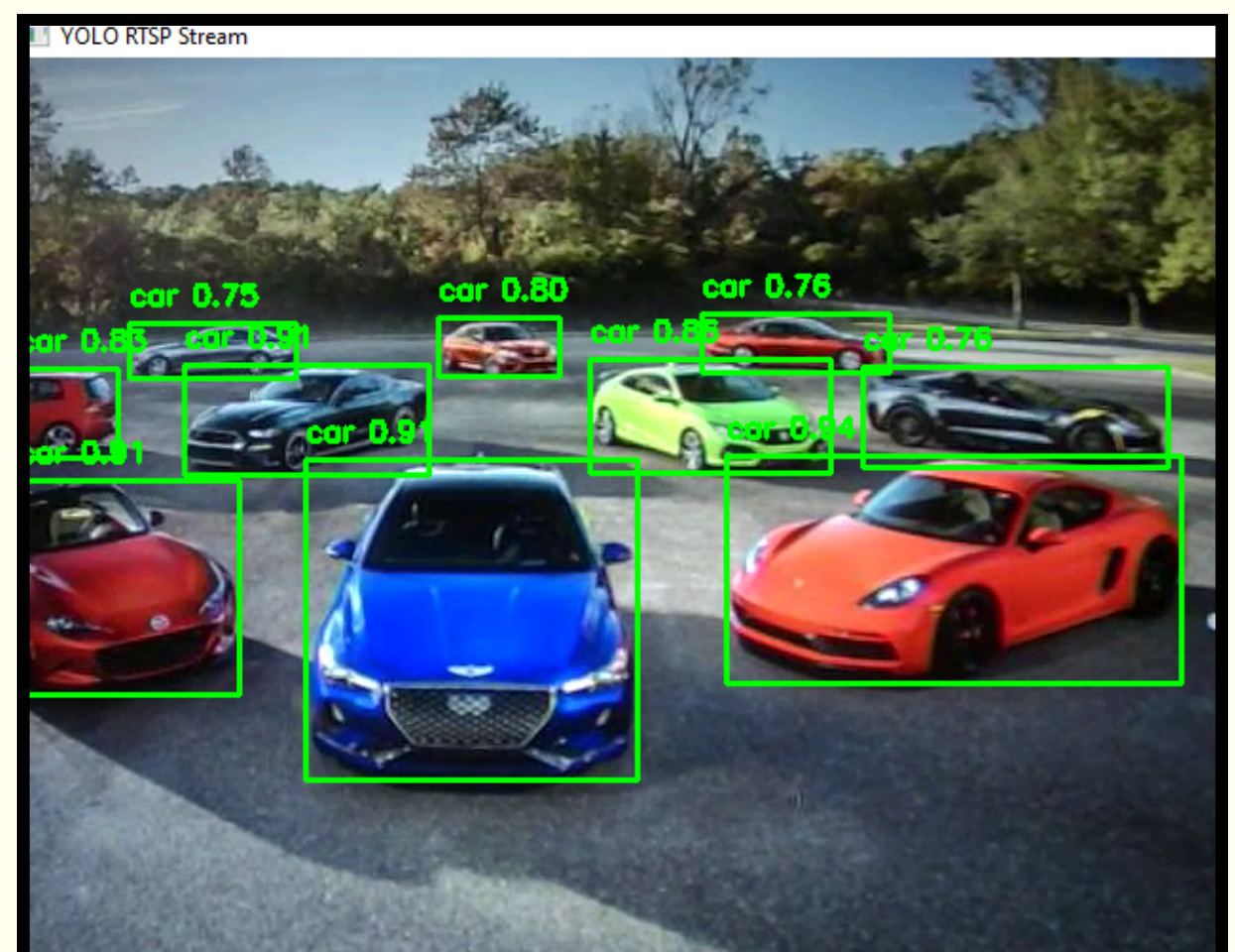
Our design prevents this from happening by utilizing features mentioned above to monitor the blind spot automatically and allow the cyclist to warn the driver.

1 COMPUTER VISION

Objective: Identify cars in cyclist's blind-spot

Components: Phone, webcam, webcam holder

- Webcam can be adjusted 90°
- Webcam monitors the blind-spot
- YOLO CV model detects approaching vehicles and sends signal to phone

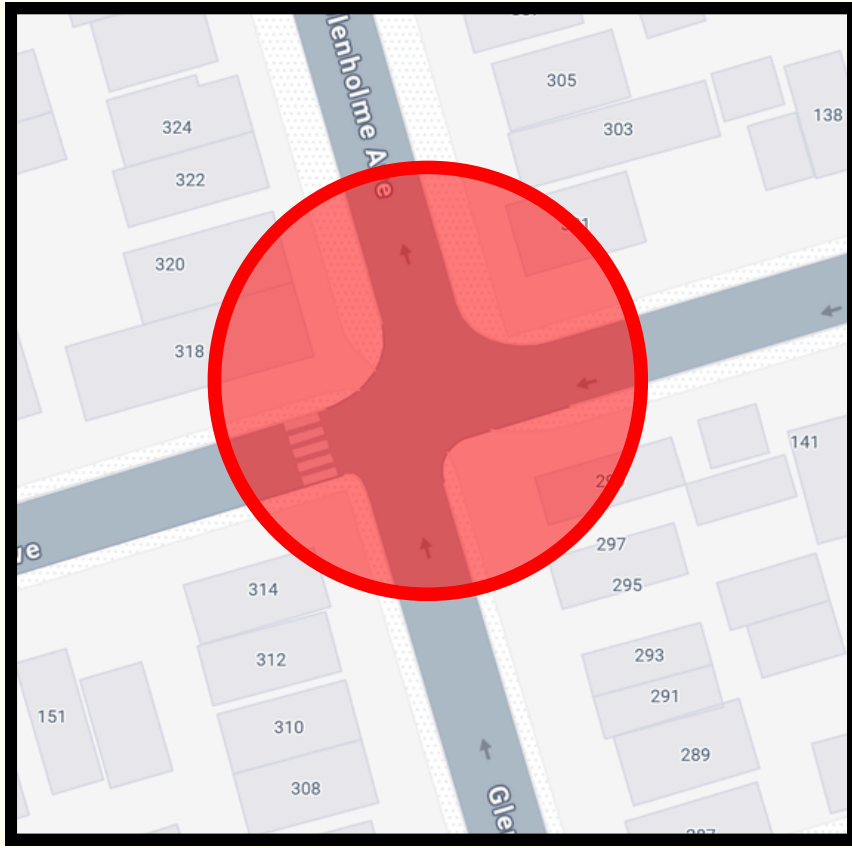


GPS SYSTEM

Objective: Detecting intersections

Components: Integrated software, Phone

- Software contains a dataset of intersection locations
- Calls GPS API (e.g. *Google Maps*) to track user's general position and turn on computer vision system when in intersection zones



3

ALERT SYSTEM

Objective: Alerting the biker via sound, light

Components: Phone, phone mount

- When at an intersection and an approaching car is detected, a buzzer sounds and flasher lights up
- Alert is sent locally on the phone

4

ELECTRIC HORN

Objective: Sending a signal to alert the driver

Components: Electric Horn, battery

- On activation, biker can choose to respond with the electrical horn
- Button and horn sound are adjustable
- Loud (90-110 dB), recognizable car horn sound

DESIGN VALUES AND DFXS

Convenience

- Minimal user input
- CV and GPS software runs automatically
- No wifi necessary
- Phone-based
- Button is adjustable for the user's hand positioning preference

Safety

- Incorporates factors of safety, design can detect cars up to 15 m away.
- Light and sound comply to existing laws
- Alerts are effective but not disruptive or hazardous

Accessibility

- Add-on to existing bikes, not requiring heavy modifications
- Components are readily available, easy to implement
- Mounts use standard sizes and adjustable

NEXT STEPS

Wireless Cameras

- Implement a wireless camera instead of a webcam
- Weather and dust-proof
- Reduce hazards from tangling wires
- Requires compatibility with phone system



External Alert System

- Send signals to external light and phone
- Develop other haptic devices for greater accessibility
- Reduce phone battery drain
- Requires edits to the app to external environment.

